

Aspect-Based Sentiment Analysis of Student Feedback Using Transformer and Large Language Models: A Comparative Study of Multidomain and Cross-Domain Learning

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Abstract—Educational institutions gather large volumes of free-text student feedback, yet manual analysis of this data is error-prone, time-consuming, and difficult to scale. This study evaluates Aspect-Based Sentiment Analysis (ABSA) on student educational reviews using five model configurations: ModernBERT, BERTweet, LLaMA 3B, LLaMA 3B with LoRA adaptation, and Gemini (zero-shot and few-shot). Models are compared across Course, Teacher, and University feedback domains under both multidomain and cross-domain training conditions. LLaMA 3B achieved the highest multidomain weighted F1-score of 0.891, with BERTweet effectively tied at 0.890; BERTweet recorded the strongest Neutral F1 (0.804), making it the preferred choice for practical deployment. Multidomain training clearly outperformed cross-domain learning across every evaluated architecture, confirming that domain-diverse training improves both accuracy and robustness in educational sentiment classification.

Keywords—Sentiment Analysis, ABSA, student feedback, feedback loop, ModernBERT, BERTweet, LLaMA 3B, LLaMA 3B with LoRA adaptation, Gemini, multidomain learning, cross-domain learning.

I. INTRODUCTION

Most educational institutions underutilize the student feedback they collect each semester [1], [2]. Many still rely on end-of-semester numerical ratings that compress nuanced opinions into a single score, discarding the aspect-level detail that free-text comments provide. Aspect-Based Sentiment Analysis (ABSA) addresses this directly by identifying specific opinion targets — such as workload, instructor clarity, or assessment difficulty — together with the sentiment expressed toward each one [3], [4], [5].

Interest in educational ABSA has grown alongside progress in NLP. Early rule-based and statistical methods showed limited effectiveness and demanded heavy manual feature engineering. Deep learning with attention mechanisms made free-text aspect extraction considerably more tractable [6], [7], and pre-trained transformers such as BERT further advanced performance [8], [9]. Most recently, instruction-tuned LLMs including GPT-style systems, LLaMA, and Gemini have enabled zero-shot and few-shot ABSA, reducing the dependence on large annotated datasets [10], [11].

Several important gaps remain. First, domain-specific ABSA covering multidomain and cross-domain settings is sparse; most studies train and test on a single feedback type, and no published work had directly compared the two strategies in educational settings [12]. Second, there is no accepted benchmark model for educational ABSA, and most studies evaluate only one or two architectures on a single dataset [13]. Third, the effect of fine-tuning on transformer and LLM performance in this domain is insufficiently studied [14]. Fourth, educational ABSA has received far less attention than e-commerce or restaurant review research [13]. To the best of our knowledge, no published study had evaluated Google Gemini for educational ABSA before this work. This paper therefore addresses four research questions:

- 1) RQ1: How does multidomain learning compare with cross-domain learning for educational ABSA?
- 2) RQ2: Can LLMs compete with fine-tuned transformer encoders?
- 3) RQ3: Which architecture performs best on educational ABSA?
- 4) RQ4: Do the trained models maintain reasonable accuracy on new, genuinely unseen student feedback?

The main contributions are:

- 1) **Dataset:** the source corpus was cleaned, normalized, and restructured at the aspect level; domain-size imbalance was corrected through oversampling and undersampling across the Course, Teacher, and University domains
- 2) **Model-based evaluation:** multidomain and cross-domain strategies were compared systematically across five model configurations
- 3) **Do main-based evaluation:** per-domain performance was computed and compared across all five models
- 4) **Error analysis:** a disagreement and winner analysis identifies where and why models differ

II. RELATED WORK

This section traces the development of ABSA methodology from early machine learning systems through deep learning, pre-trained transformers, and generative LLMs, before examining ABSA in educational settings and identifying the research gaps that motivate this study.

A. Traditional Machine Learning Approaches

The earliest ABSA systems relied on rule-based heuristics and conventional classifiers — Naïve Bayes, Support Vector Machines, Decision Trees, and Random Forests — using bag-of-words text representations [15]. Their principal weakness is dependence on manually constructed feature sets and fixed aspect categories, which generalize poorly across institutional contexts [16]. For short, topically focused text such as student feedback, Naïve Bayes can capture sentiment patterns efficiently without the overfitting risk that more complex classifiers sometimes introduce [16], [17].

B. Deep Learning and Attention-Based Methods

Recurrent networks with attention mechanisms marked the next step forward, learning relationships between aspect words and surrounding text directly from data. Target-dependent LSTMs conditioned sentiment prediction on the target aspect, producing more accurate results without relying on external sentiment lexicons or parse trees [18]. Building on that idea, embedding aspect information into the attention mechanism allowed the model to weight words according to their relevance to the aspect being analysed [6]. Within education, an ALBERT-BiLSTM [36] model with attention achieved 91.6% accuracy on MOOC course reviews, outperforming standard LSTM (80.1%), BiLSTM (81.9%), and CNN (81.1%) models — demonstrating the gains available when pre-trained language representations are combined with bidirectional context modelling [19]. The main advantage of deep learning is automatic feature discovery, eliminating manual engineering [20]; the trade-off is a requirement for larger labelled datasets, greater computational resources, and longer training times, all of which are constraints in resource-limited educational settings [21].

C. Transformer-Based and Pre-Trained Language Models

BERT marked a step change in NLP by generating deeply contextualized word representations through bidirectional pre-training, allowing the model to understand word meaning from both preceding and following context simultaneously [8]. Sun et al. [22] adapted BERT for ABSA by framing the task as a question-answering problem, where the target aspect is posed as the question and the review text is the passage. In a comparison of five transformer architectures on 3,500 annotated Trustpilot reviews covering 14 e-commerce aspects, RoBERTa achieved the best F1-score of 92.53%, ahead of BERT (90.12%) and DistilBERT (88.47%) [23]. DistilBERT is attractive in resource-constrained settings because it has roughly 40% fewer parameters than BERT while maintaining competitive accuracy, reaching 87% F1 on a customer feedback benchmark [24]. Within education, fine-tuned BERT-based models have consistently reported F1-scores between 85% and 92%, though typically on datasets of only around 2,000 annotated reviews — a data-scarcity challenge discussed further in Section II-E.

D. Large Language Models and Generative Approaches

Instruction-tuned LLMs, trained using reinforcement learning from human feedback, have changed the approach to ABSA. Rather than requiring large labelled datasets, these models identify aspects and sentiments from natural-language instructions in zero-shot or few-shot settings, which suits resource-constrained educational deployments. Shah et al. [10] evaluated GPT-4, GPT-3.5-turbo, and LLaMA-2 (7B–70B) on app-review feature-sentiment extraction, finding that every LLM outperformed a rule-based baseline in the zero-shot setting, with GPT-4 exceeding the best baseline by 23.6 percentage points. Providing five in-context examples raised GPT-4’s F1-score by a further six points to 56.1% [10]. Even so, GPT-4 performed considerably better on positive reviews (76% F1) than on neutral ones (45% F1), and a fine-tuned REBERT model still outperformed GPT-4 by six points, indicating that supervised fine-tuning remains superior when labelled data are available [10]. In a lower-resource context, Fu et al. [25] found that GPT-3.5 and GPT-4 reached 92.1% and 95.3% accuracy on Cantonese counselling transcripts, outperforming LSTM and SVM baselines. Fauzan [26] applied zero-shot DeBERTa-v3 and DistilBERT to roughly 17,500 tweets about AI chatbots and obtained only 38% aspect-detection accuracy, showing that zero-shot ABSA scales well but still has precision limits. The Gemini team reported that Gemini Ultra exceeded GPT-4 on several benchmarks [11], but no published IEEE or ScienceDirect study had evaluated Gemini for educational ABSA before this work. Finally, Touvron et al. [27] and Grattafiori et al. [28] demonstrated that open-weight LLaMA models carry an additional advantage for educational institutions because they can be deployed on local servers, ensuring student feedback never leaves institutional infrastructure while still delivering competitive few-shot performance.

E. ABSA in Educational Contexts and Dataset Scarcity

Applying ABSA to student feedback is harder than applying it to e-commerce or service reviews. Student comments are typically informal, often hedge criticism, and frequently contain implicit aspect references: a comment such as “I wish there were more practice problems” signals dissatisfaction with assessment design without naming that aspect explicitly; a single review may simultaneously praise the instructor while criticizing the course textbook. Most previous work has focused on MOOC-style course reviews [19], and Schouten and Frasnar [29] showed that student feedback relies more heavily on personal pronouns and implicit topics than product reviews do, a finding corroborated by more recent cross-domain ABSA research [12]. This characteristic means models pre-trained on restaurant or laptop review corpora frequently perform poorly when applied to educational feedback without domain-specific fine-tuning, as shown by García-Pablos et al. [34] and confirmed by later studies on domain adaptation in ABSA [31]. A further obstacle is the scarcity of labelled educational data: most studies draw on fewer than 2,000

annotated examples, pushing researchers toward general-purpose models rather than rigorously fine-tuned ones.

F. Summary of Research Gaps

- Dataset scarcity:** most educational ABSA studies use fewer than 2,000 samples, limiting model development and fair comparison
- Implicit aspect extraction:** current models typically require explicitly stated aspects, while students often express opinions without naming the target
- Joint extraction and classification:** most educational studies assume pre-defined aspects; joint extraction-classification has shown benefits elsewhere but has not been rigorously evaluated on educational data
- Cross-institution generalization:** models trained at one institution may not transfer to others due to differences in terminology, feedback instruments, and student demographics
- Explainability:** deep models behave as black boxes; administrators need to understand why a sentiment label was assigned before acting on results
- Gemini for ABSA:** no published study had evaluated Google Gemini for educational ABSA before this work
- LLM hallucination:** LLMs can confuse aspects and misclassify neutral sentiment as negative, and these failure modes had not been characterized specifically for educational feedback

This study directly targets gaps 1, 3, 5, 6, and 7 by constructing a balanced 12,654-sample multidomain dataset, comparing multidomain against cross-domain generalization, providing the first educational evaluation of Gemini, and conducting a detailed disagreement and failure-mode analysis.

III. METHODOLOGY

This section describes the full experimental pipeline: dataset construction and preprocessing, input formatting, model selection, the Gemini prompting strategy, the experimental design, hyperparameter tuning, and evaluation metrics.

A. Dataset and Pre-Processing

The dataset was drawn from the ASQE educational review corpus, a publicly available, aspect-annotated collection of student feedback. Training, validation, and test files from one partition were merged into a single pool. Each record contained one or more annotated aspect-sentiment pairs, each specifying an aspect term, a sentiment polarity (Positive, Negative, or Neutral), and an aspect category label. For cross-domain evaluation, each review was assigned to one of three domains — Course, Teacher, or University — based on the prefix of its aspect category label, with a fixed priority of University > Teacher > Course for reviews spanning multiple categories. Because one review may contain multiple annotated aspects, the dataset was converted from review-level to aspect-level format, creating one training instance per valid aspect-sentiment pair with four fields: review text, aspect term, sentiment label, and

domain. A multi-stage cleaning pipeline removed invalid entries, deduplicated aspect annotations within the same review, discarded noisy aspect terms (single characters, pronouns, phrases exceeding four words), and standardized frequently recurring expressions.

After cleaning, the corpus contained 4,218 Course, 5,576 Teacher, and 1,175 University aspect-level samples. To correct the domain-size imbalance, the University domain was expanded by random oversampling and the Teacher domain reduced by random undersampling, both using the Course domain (4,218 samples) as the reference size, yielding a balanced dataset of 12,654 samples — exactly 4,218 per domain. This was split 80/10/10 (train/validation/test) using stratified sampling to preserve the sentiment class distribution. Table I summarizes the dataset statistics.

Table I. Dataset Statistics Before and After Balancing

Domain	Before Balancing	After Balancing
Course	4,218	4,218
Teacher	5,576	4,218
University	1,175	4,218
Total	10,969	12,654

B. Input Formatting

All supervised models received a structured input string that makes domain and aspect information explicit at the token level:

[DOMAIN] {domain} [ASPECT] {aspect} [TEXT] {review}

For example: [DOMAIN] university [ASPECT] community [TEXT] Exeter University was my first choice... All inputs were tokenized using each model’s pre-trained tokenizer, padded or truncated to a maximum of 128 tokens.

C. Models Evaluated

Five configurations were selected to enable a systematic comparison across fine-tuned encoder-based transformers and LLM-based approaches [10]. ModernBERT [32] (answerdotai/ModernBERT-base) is a recently introduced bidirectional encoder optimized for long-context understanding and memory-efficient inference. BERTweet [33] (vinai/bertweet-base), pre-trained on 850 million English tweets, is well suited to the short, informal language of student feedback; its tokenizer was loaded with normalization enabled. LLaMA 3B [28] was fine-tuned end-to-end under supervised training for direct comparison with the transformer encoders. LLaMA 3B with LoRA [35] reduced trainable parameters by decomposing weight updates into low-rank matrices, substantially cutting computational cost while retaining competitive performance. Gemini [11] (gemini-2.5-flash) was evaluated through prompt engineering only, without supervised fine-tuning. Table II lists the hardware platform used for each model.

Table II. Hardware Platform per Model

Model	Platform
ModernBERT	Google Colab
BERTweet	Google Colab
Gemini	Google Colab
LLaMA 3B	Renku
LLaMA 3B + LoRA	Renku

D. Prompting Strategy for Gemini

Since Gemini was not fine-tuned, its evaluation relied entirely on prompt design. For zero-shot evaluation:

“Classify the sentiment of the aspect in the review. Review: {review}. Aspect: {aspect}. Answer ONLY one word: positive, negative, or neutral.”

For few-shot evaluation, six labelled examples covering all three sentiment classes across multiple educational domains were prepended, together with the system instruction: “You are an expert Aspect-Based Sentiment Analysis classifier. Determine the sentiment of the given aspect.” A retry mechanism with up to five attempts per instance and a five-second inter-retry delay handled API rate-limit errors; where all retries failed, the prediction defaulted to neutral.

E. Experimental Design

Two experiments assessed domain generalization. In the multidomain experiment, all supervised models were trained on the full combined balanced dataset and evaluated on a shared held-out test set from all three domains. In the cross-domain experiment, models were trained on two domains and tested on the entirely unseen third, yielding three splits: Course + Teacher → University, Course + University → Teacher, and Teacher + University → Course. Each model was retrained from scratch for every split to prevent information leakage. Gemini was prompted independently on each test domain without domain-combination training.

F. Hyperparameter Tuning

All supervised models were trained using the HuggingFace Trainer API with: batch size 16, weight decay 0.01, mixed-precision (fp16) training, and class-weighted cross-entropy loss computed per batch using sklearn’s compute_class_weight with the balanced strategy. Learning rate was varied across {1e-5, 2e-5, 3e-5, 5e-5} and training epochs across {3, 5, 7}, with validation at the end of each epoch.

G. Evaluation Metrics

Performance was quantified using accuracy, macro and weighted precision/recall/F1-score, and the class-specific Neutral F1-score, consistent with standard ABSA evaluation practice [4], [14]. Macro metrics treat all classes equally regardless of support, making them sensitive to Neutral-class performance. Weighted metrics account for class support. Neutral F1-score is reported separately given the consistent difficulty all models experienced with that class. Weighted F1-score serves as the primary ranking metric throughout.

IV. PERFORMANCE EVALUATION

All results reflect sentiment classification on pre-annotated aspect terms: the aspect for each review was already identified in the dataset, and the model’s task was to predict the correct sentiment polarity (Positive, Negative, or Neutral) for that aspect. No aspect-extraction step was evaluated — a distinction revisited in the conclusion.

A. Multidomain Results

Table III reports the complete hyperparameter sweep for all four supervised architectures in the multidomain setting.

Table III. Multidomain Tuning Results

Model	Category	Configuration	Acc	Mac-F1	W-F1	Neu-F1
ModernBERT	Transformer	LR=1e-5	0.858	0.768	0.861	0.530
ModernBERT	Transformer	LR=2e-5	0.878	0.799	0.880	0.591
ModernBERT	Transformer	4 Epochs	0.883	0.798	0.882	0.577
ModernBERT	Transformer	LR=3e-5	0.883	0.806	0.884	0.600
ModernBERT	Transformer	No Wt. Loss	0.883	0.791	0.879	0.560
BERTweet	Transformer	Baseline	0.876	0.800	0.882	0.766
BERTweet	Transformer	LR=3e-5	0.878	0.796	0.879	0.771
BERTweet	Transformer	LR=1e-5	0.869	0.787	0.870	0.722
BERTweet	Transformer	Batch=32	0.878	0.800	0.882	0.777
BERTweet	Transformer	Batch=8	0.883	0.808	0.884	0.783
BERTweet	Transformer	Epoch=3	0.885	0.801	0.886	0.795
BERTweet	Transformer	Epoch=7	0.884	0.799	0.890	0.804
LLaMA 3B	LLM-Based	Original (no FT)	0.646	0.583	0.698	0.307
LLaMA 3B	LLM-Based	LR=1e-5	0.852	0.703	0.844	0.340
LLaMA 3B	LLM-Based	LR=5e-5, Ep=3	0.887	0.817	0.891	0.617
LLaMA 3B	LLM-Based	LR=5e-5, Ep=5	0.888	0.820	0.891	0.627
LoRA (LLaMA 3B)	LLM-Based	Baseline	0.740	0.616	0.760	0.363
LoRA (LLaMA 3B)	LLM-Based	LR=3e-5	0.818	0.719	0.827	0.520
LoRA (LLaMA 3B)	LLM-Based	LR=5e-5, Ep=5	0.852	0.755	0.860	0.520
LoRA (LLaMA 3B)	LLM-Based	LR=5e-5, Ep=7	0.856	0.757	0.862	0.493

Table IV. Cross-Domain Results — LLM-Based Models

Model	Heldout Domain	Configuration	Acc	Mac-F1	W-F1	Neu-F1
LLaMA 3B	Course	5 Epochs	0.742	0.671	0.733	0.42
LLaMA 3B	Course	3 Epochs	0.539	0.477	0.517	0.27
LLaMA 3B	Course	LR=3e-5	0.335	0.287	0.267	0.23
LLaMA 3B	Teacher	3 Epochs	0.859	0.760	0.857	0.48
LLaMA 3B	Teacher	5 Epochs	0.853	0.729	0.847	0.39
LLaMA 3B	Teacher	LR=3e-5	0.608	0.440	0.587	0.04
LLaMA 3B	University	3 Epochs	0.922	0.736	0.928	0.36
LLaMA 3B	University	5 Epochs	0.934	0.747	0.934	0.40
LLaMA 3B	University	LR=3e-5	0.723	0.477	0.752	0.19
LoRA (LLaMA 3B)	Course	5 Epochs	0.746	0.652	0.724	0.37
LoRA (LLaMA 3B)	Course	3 Epochs	0.731	0.609	0.696	0.27
LoRA (LLaMA 3B)	Course	LR=3e-5	0.748	0.654	0.725	0.39
LoRA (LLaMA 3B)	Teacher	3 Epochs	0.765	0.635	0.755	0.32

LoRA (LLaMA 3B)	Teacher	5 Epochs	0.774	0.654	0.764	0.37
LoRA (LLaMA 3B)	Teacher	LR=3e-5	0.777	0.661	0.767	0.38
LoRA (LLaMA 3B)	University	3 Epochs	0.896	0.677	0.882	0.40
LoRA (LLaMA 3B)	University	5 Epochs	0.890	0.665	0.877	0.37
LoRA (LLaMA 3B)	University	LR=3e-5	0.895	0.672	0.880	0.38

Table V. Cross-Domain Results — Transformer-Based Models

Model	Heldout Domain	Configuration	Acc	Mac-F1	W-F1	Neu-F1
ModernBERT	Course	5 Epochs	0.761	0.680	0.751	0.397
ModernBERT	Course	3 Epochs	0.864	0.754	0.877	0.473
ModernBERT	Course	LR=3e-5	0.795	0.725	0.788	0.473
ModernBERT	Teacher	5 Epochs	0.822	0.710	0.822	0.398
ModernBERT	Teacher	3 Epochs	0.828	0.722	0.838	0.465
ModernBERT	Teacher	LR=3e-5	0.848	0.746	0.848	0.465
ModernBERT	University	5 Epochs	0.912	0.714	0.917	0.353
ModernBERT	University	3 Epochs	0.763	0.710	0.779	0.353
ModernBERT	University	LR=3e-5	0.909	0.686	0.914	0.259
BERTweet	Course	3 Epochs	0.749	0.691	0.754	0.432
BERTweet	Course	5 Epochs	0.784	0.713	0.780	0.449
BERTweet	Course	LR=3e-5	0.778	0.696	0.767	0.416
BERTweet	Teacher	3 Epochs	0.818	0.699	0.815	0.374
BERTweet	Teacher	5 Epochs	0.835	0.707	0.829	0.366
BERTweet	Teacher	LR=3e-5	0.845	0.734	0.840	0.443
BERTweet	University	3 Epochs	0.925	0.710	0.929	0.313
BERTweet	University	5 Epochs	0.933	0.740	0.936	0.386
BERTweet	University	LR=3e-5	0.916	0.710	0.919	0.321

LLaMA 3B and BERTweet recorded near-identical weighted F1-scores (0.891 and 0.890 respectively) — a gap that is not practically meaningful. Beneath that aggregate similarity lies an important difference: BERTweet handled neutral comments far better, achieving a Neutral F1-score of 0.804 against LLaMA’s 0.627. This likely reflects BERTweet’s Twitter pre-training, which exposes it to the short, informal, and often ambiguous language that closely mirrors how students express neutral feedback. ModernBERT followed closely with a weighted F1-score of 0.884. LoRA-adapted LLaMA reached 0.862 — the lowest among fine-tuned models but well above the untuned LLaMA baseline (0.698) — confirming that both careful hyperparameter tuning and parameter-efficient fine-tuning can meaningfully improve performance.

B. Cross-Domain Results

Tables IV and V report detailed cross-domain results across all model configurations and all three held-out domain splits. Table IV covers LLM-based models (LLaMA 3B and LoRA), while Table V covers transformer-based models (ModernBERT and BERTweet). Looking across both tables, BERTweet and LLaMA 3B achieved essentially the same top weighted F1-score on the University domain (0.936 and 0.934, respectively), making them effectively that full fine-tuning of large models can be unstable on smaller or harder domains. LoRA-adapted LLaMA, by contrast, stayed far more consistent across all three domains

and configurations, never dropping below 0.696, suggesting that parameter-efficient fine-tuning acts as a form of regularization. In terms of Macro F1 and Neutral F1, the best results came from LLaMA 3B (full fine-tuning) on the Teacher domain with 3 epochs (Macro F1 = 0.760, Neutral F1 = 0.48), the highest Neutral F1 across both tables. This indicates that while the University domain gave the highest overall weighted F1-scores, Teacher-domain data allowed the model to balance all three sentiment classes more fairly rather than being dominated by the majority class.

Table VI quantifies the performance drop when moving from each model’s best multidomain result to its average across all three cross-domain splits.

Table VI. Multidomain vs. Cross-Domain Performance Comparison

Model	Best Multi WF1	Avg Cross WF1	Δ (Multi – Cross)
ModernBERT	0.884	0.881	+0.003
BERTweet	0.890	0.852	+0.038
LLaMA 3B	0.891	0.841	+0.050
LoRA	0.862	0.791	+0.071

ModernBERT showed the most stable behaviour (+0.003 gap), suggesting it is largely domain-agnostic. BERTweet displayed a moderate drop (+0.038), LLaMA’s decline was somewhat larger (+0.050), and LoRA experienced the greatest decrease (+0.071), consistent with parameter-efficient fine-tuning making the model more sensitive to the domains seen during training. Across all four architectures, the University domain was the easiest to transfer to and the Course domain was the hardest — most likely because course-related feedback contains a wider range of subject-specific vocabulary.

C. Gemini Results (Zero-Shot and Few-Shot)

Gemini was evaluated through prompt engineering rather than supervised fine-tuning and should not be compared directly with the fine-tuned models. Rather than being trained on domain pairs, Gemini was prompted independently on each domain, so its results reflect how well prompting alone adapts to different educational feedback types. Table VII reports Weighted F1 and Neutral F1 for each condition (accuracy values are discussed in prose).

Table VII. Gemini Zero-Shot (ZS) and Few-Shot (FS) Performance

Domain	ZS-WF1	ZS-NF1	FS-WF1	FS-NF1
Course	0.76	0.28	0.76	0.29
Teacher	0.84	0.25	0.84	0.34
University	0.95	0.36	0.94	0.35

Gemini performed strongly on the University domain (ZS Accuracy = 0.95, WF1 = 0.95; FS Accuracy = 0.94, WF1 = 0.94) with no task-specific training, while Teacher and Course domains were lower but solid (WF1 = 0.84 and 0.76, respectively). Neutral sentiment was the consistent weakness: Neutral F1-scores ranged from only 0.25 to 0.36 — well below every fine-tuned model, including LoRA

(Neutral F1 = 0.493). Few-shot prompting improved Neutral F1 modestly for the Teacher domain (0.25 → 0.34) and marginally for Course (0.28 → 0.29), but slightly reduced it on University (0.36 → 0.35), suggesting the additional examples introduced minor noise in a domain Gemini already handled well.

D. Winner and Disagreement Analysis

To understand why multidomain training generally outperformed cross-domain training, predictions from each model’s multidomain and cross-domain versions were compared directly on the same test instances. Table VIII records how often the multidomain model was correct while the cross-domain model was wrong (“Multidomain Wins”) and the reverse.

Table VIII. Winner Analysis: Multidomain vs. Cross-Domain

Model	Multi Wins	Cross Wins	Net Adv.
ModernBERT	181	89	+92
BERTweet	322	19	+303
LLaMA 3B	139	51	+88
LoRA	225	136	+89

BERTweet showed the largest net advantage for multidomain training (322 vs. 19), indicating that it benefits substantially from exposure to diverse domains during training. Table IX reports the disagreement rate across the full 2,532-sample test set.

Table IX. Disagreement Analysis Across the Full Test Set (n = 2,532)

Model	Samples	Disagree.	Rate
ModernBERT	2,532	306	12.1%
BERTweet	2,532	356	14.1%
LLaMA 3B	2,532	424	16.7%
LoRA	2,532	487	19.2%

ModernBERT’s disagreement rate (12.1%) is the smallest of the four, consistent with its near-identical multidomain and cross-domain performance and confirming that it generalizes well regardless of training configuration. LoRA showed the highest disagreement rate (19.2%), indicating the greatest sensitivity to domain variation — though the gap between its multidomain wins (225) and cross-domain wins (136) was narrower than for other models, suggesting its cross-domain version can still make competitive predictions at a much lower training cost.

E. Neutral Sentiment Analysis

Neutral sentiment was the most difficult class for every model. Unlike clearly positive or negative comments, neutral feedback often uses vague, hedged, or mixed language that overlaps with the vocabulary of genuinely negative statements. Shah et al. [10] reported the same pattern for GPT-4 on app reviews, attributing it to the inherent ambiguity of neutral language. Among supervised models, BERTweet achieved the strongest Neutral F1 (0.804), followed by LLaMA 3B (0.627), ModernBERT(0.600), and LoRA (0.493). Gemini performed considerably worse on neutral feedback, with Neutral F1-scores of only 0.25–0.36 across all domains and prompting strategies, reinforcing that supervised fine-tuning remains far more effective than prompting when detecting subtle sentiment ambiguity.

F. Confusion Matrix Analysis

Confusion matrices were examined on the University domain, where all models achieved their best overall performance (Tables X–XIII). The main diagonal dominates in every case, indicating correct classification for the large majority of instances. The Neutral class was the most error-prone throughout, most often misclassified as Negative rather than Positive — consistent with the Neutral analysis above. LLaMA 3B produced the fewest total errors, matching its highest domain-level WF1.

Table X. Confusion Matrix — LLaMA 3B on University Domain

Act. \ Pred.	Positive	Neutral	Negative
Positive	673	0	3
Neutral	0	28	1
Negative	2	0	825

Table XI. Confusion Matrix — BERTweet on University Domain

Act. \ Pred.	Positive	Neutral	Negative
Positive	668	2	8
Neutral	1	27	1
Negative	3	0	824

Table XII. Confusion Matrix — ModernBERT on University Domain

Act. \ Pred.	Positive	Neutral	Negative
Positive	668	2	8
Neutral	1	27	1
Negative	3	0	824

Table XIII. Confusion Matrix — LoRA (LLaMA 3B) on University Domain

Act. \ Pred.	Positive	Neutral	Negative
Positive	669	1	8
Neutral	1	26	2
Negative	1	1	824

G. Research Questions Addressed

RQ1: Multidomain learning outperformed cross-domain learning for every architecture, though the gap varied. ModernBERT was virtually unaffected by training configuration (+0.003), suggesting inherently strong domain-agnostic properties, while LLaMA and LoRA benefited more substantially (+0.050 and +0.071). Training across multiple domains is generally beneficial, but its practical importance depends on the model architecture.

RQ2: Yes. LLaMA 3B achieved the highest aggregate weighted F1 (0.891), though its margin over BERTweet (0.890) is negligible. LoRA reached 0.862 at considerably lower computational cost. Gemini achieved WF1 = 0.95 on the University domain through prompting alone, providing compelling evidence that modern LLMs can be competitive on educational sentiment tasks without fine-tuning.

RQ3: The result is more nuanced than a single top-ranked model. LLaMA 3B leads on aggregate weighted F1, but the margin over BERTweet is 0.001 and is not practically meaningful. BERTweet substantially outperforms LLaMA on Neutral sentiment (0.804 vs. 0.627) — the most consequential class in student feedback, where misclassification misleads institutional decision-making. For real-world deployment, BERTweet is the stronger practical choice.

RQ4: The cross-domain and unseen-dataset evaluations (Section IV-I) together provide a clear answer. ModernBERT proved the most stable across unseen feedback types, with the highest accuracy on the external corpus (76.7%). BERTweet, despite ranking first on the in-distribution test set, dropped sharply on genuinely unseen data. Full validation on feedback from genuinely different institutions remains necessary before broad deployment.

H. Final Ranking

Based on weighted F1-score, LLaMA 3B and BERTweet are effectively tied as top performers. BERTweet’s clear advantage on Neutral sentiment detection makes it the preferred choice for practical deployment. Table XIV presents the final ranking.

Table XIV. Final Model Ranking

Model	W-F1	Neu-F1
LLaMA3B (1st)	0.891	0.627
BERTweet (2nd)	0.890	0.804
ModernBERT (3rd)	0.884	0.600
LoRA-LLaMA (4th)	0.862	0.493
Gemini FS (Univ.)	0.94	0.35

I. Validation on Genuinely Unseen Educational Feedback

To test real-world generalization, the four available models were evaluated on a separate educational feedback corpus with no overlap with the training or test data used elsewhere in this study. Table XV reports the results. ModernBERT produced the strongest performance (Accuracy = 76.7%, Macro F1 = 0.768, Neutral F1 = 0.685), confirming that its domain-agnostic encoder properties — already visible in its near-zero cross-domain drop (+0.003) — translate into the best transfer when the target distribution is entirely novel. BERTweet fell to last place (Accuracy = 34.9%, Macro F1 = 0.287), a stark reversal of its position on the original test set. This suggests that BERTweet’s Twitter pre-training is a distribution-specific advantage: it closely mirrors the informal, abbreviated language in the training corpus but does not transfer as well to educational feedback written in a different register. Gemini Zero-shot ranked second (Accuracy = 51.6%, Macro F1 = 0.482) despite receiving no task-specific training, and both Gemini conditions substantially outperformed BERTweet on this out-of-distribution data. These results revise the ranking from Section IV-H in an important way: for deployment on feedback from institutions or formats not seen during training, ModernBERT is the most reliable choice, not BERTweet. High performance on an in-distribution test set does not guarantee robust generalization, and model selection for educational ABSA should account for the likely degree of domain shift in the target deployment environment.

Table XV. Performance on Genuinely Unseen Educational Feedback

Model	Acc	Mac-P	Mac-F1	WF1	NF1
ModernBERT	0.767	0.782	0.768	0.764	0.685
Gemini (ZS)	0.516	0.461	0.482	0.465	0.142

Gemini (FS)	0.420	0.447	0.433	0.428	0.291
BERTweet	0.349	0.428	0.287	0.282	0.173

J. Threats to Validity

- Dataset scope:** the corpus covers three educational feedback types from a limited number of institutions; differences in terminology, rating culture, and student demographics may limit generalizability
- Model scale:** computational constraints limited the LLM evaluation to LLaMA 3B; larger models such as LLaMA 3-70B may yield different results, particularly on the Neutral class
- Gemini comparability:** because Gemini was evaluated via prompting rather than fine-tuning, its results are not directly comparable with those of the other architectures; prompt reformulation or a newer version could alter performance
- Pre-annotated aspects:** all models received the aspect term in advance; in real deployment, an upstream aspect-extraction stage would add additional errors.

V. CONCLUSION AND FUTURE WORK

This paper compared transformer encoders (ModernBERT, BERTweet) and LLM-based configurations (LLaMA 3B, LLaMA 3B with LoRA, Gemini) on student feedback sentiment classification under both multidomain and cross-domain training conditions. The central finding is clear: multidomain training outperformed cross-domain transfer for every model tested. ModernBERT was the most stable across settings, while LLaMA 3B and BERTweet finished in a practical tie on aggregate performance (0.891 vs. 0.890 weighted F1). BERTweet’s substantially higher Neutral F1-score (0.804 vs. 0.627) makes it the more appropriate choice for real-world educational deployment, since neutral comments are both the most common source of ambiguity and the class where misclassification is most likely to mislead institutional decision-making. Gemini performed well without fine-tuning, reaching 0.95 weighted F1 on the University domain through prompting alone, but it struggled with Neutral sentiment just as every other model did. The consistent Neutral-class weakness across all five configurations, combined with the reliance on pre-annotated aspect terms rather than automated extraction, are the principal limitations of this work.

Future work should prioritize automating aspect extraction so the full pipeline can operate on raw text. Testing larger models — LLaMA 3-70B, GPT-4-class systems, or updated Gemini versions — would clarify whether the Neutral-sentiment gap is a scale issue or a deeper architectural one, while targeted techniques such as improved class balancing or dedicated neutral-detection components may address it more directly. Evaluating the pipeline on feedback from institutions not represented in training data would confirm how broadly these findings generalize, and adding interpretability features so instructors understand why a sentiment was assigned would improve trust and adoption. Ultimately, piloting a working prototype within a live course-feedback loop is the natural next step toward making this research operationally useful for educational quality assurance.

VI. GENERATIVE AI USAGE DISCLOSURE

Generative AI tools were used in this work at two distinct levels. First, Gemini 2.5 Flash (Google DeepMind) was used as an evaluated experimental model; its zero-shot and few-shot sentiment classification performance constitutes a primary research contribution and is reported in Sections IV-C and IV-I. Second, Claude (Anthropic) was used to assist with academic writing, including prose editing, paragraph restructuring, and language refinement across Sections I, II, III, IV, and V. All experimental results, data collection, model training, hyperparameter tuning, and analytical conclusions reflect the authors' own independent work; AI assistance was limited strictly to the linguistic presentation of findings already established through the authors' own research process.

REFERENCES

- [1] J. Flodén, "The impact of student feedback on teaching in higher education," *Assess. Eval. High. Educ.*, vol. 42, no. 7, pp. 1054–1068, Oct. 2017, doi: 10.1080/02602938.2016.1224997.
- [2] D. Boud, "Feedback: ensuring that it leads to enhanced learning," *Clin. Teach.*, vol. 12, no. 1, pp. 3–7, Feb. 2015, doi: 10.1111/tct.12345.
- [3] B. Liu, *Sentiment Analysis and Opinion Mining*, in *Synthesis Lectures on Human Language Technologies*. Cham, Switzerland: Springer, 2012, doi: 10.1007/978-3-031-02145-9.
- [4] P. Ren, L. Yang, and F. Luo, "Automatic scoring of student feedback for teaching evaluation based on aspect-level sentiment analysis," *Educ. Inf. Technol.*, vol. 28, no. 1, pp. 797–814, Jan. 2023, doi: 10.1007/s10639-022-11151-z.
- [5] A. Jazuli, Widowati, and R. Kusumaningrum, "Optimizing aspect-based sentiment analysis using BERT for comprehensive analysis of Indonesian student feedback," *Appl. Sci.*, vol. 15, no. 1, p. 172, Dec. 2024, doi: 10.3390/app15010172.
- [6] Y. Wang, M. Huang, X. Zhu, and L. Zhao, "Attention-based LSTM for aspect-level sentiment classification," in *Proc. 2016 Conf. Empirical Methods Natural Language Process.*, Austin, TX, USA, 2016, pp. 606–615, doi: 10.18653/v1/D16-1058.
- [7] Z. Ping, G. Sang, Z. Liu, and Y. Zhang, "Aspect category sentiment analysis based on prompt-based learning with attention mechanism," *Neurocomputing*, vol. 565, p. 126994, Jan. 2024, doi: 10.1016/j.neucom.2023.126994.
- [8] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," *arXiv:1810.04805*, 2018, doi: 10.48550/ARXIV.1810.04805.
- [9] N. M. Gardazi, A. Daud, M. K. Malik, A. Bukhari, T. Alsahfi, and B. Alshemaimri, "BERT applications in natural language processing: a review," *Artif. Intell. Rev.*, vol. 58, no. 6, p. 166, Mar. 2025, doi: 10.1007/s10462-025-11162-5.
- [10] F. A. Shah, A. Sabir, R. Sharma, and D. Pfahl, "How effectively do LLMs extract feature-sentiment pairs from app reviews?," *arXiv:2409.07162*, 2024, doi: 10.48550/ARXIV.2409.07162.
- [11] Gemini Team et al., "Gemini: A family of highly capable multimodal models," *arXiv:2312.11805*, 2023, doi: 10.48550/ARXIV.2312.11805.
- [12] F. Ouyang and B. Shen, "A mutual mean teacher framework for cross-domain aspect-based sentiment analysis," *J. Supercomput.*, vol. 80, no. 7, pp. 9073–9095, May 2024, doi: 10.1007/s11227-023-05792-1.
- [13] Y. C. Hua, P. Denny, J. Wicker, and K. Taskova, "EduRABSA: An education review dataset for aspect-based sentiment analysis tasks," *arXiv:2508.17008*, 2025, doi: 10.48550/arXiv.2508.17008.
- [14] T. Shaik et al., "A review of the trends and challenges in adopting natural language processing methods for education feedback analysis," *IEEE Access*, vol. 10, pp. 56720–56739, 2022, doi: 10.1109/ACCESS.2022.3177752.
- [15] Y. C. Hua, P. Denny, J. Wicker, and K. Taskova, "A systematic review of aspect-based sentiment analysis: domains, methods, and trends," *Artif. Intell. Rev.*, vol. 57, no. 11, p. 296, Sep. 2024, doi: 10.1007/s10462-024-10906-z.
- [16] I. A. Kandhro, F. Ali, M. Uddin, A. Kehar, and S. Manickam, "Exploring aspect-based sentiment analysis: an in-depth review of current methods and prospects for advancement," *Knowl. Inf. Syst.*, vol. 66, no. 7, pp. 3639–3669, Jul. 2024, doi: 10.1007/s10115-024-02104-8.
- [17] M. Sivakumar and U. S. Reddy, "Aspect based sentiment analysis of students opinion using machine learning techniques," in *Proc. 2017 Int. Conf. Inventive Comput. Informatics (ICICI)*, Coimbatore, India, 2017, pp. 726–731, doi: 10.1109/ICICI.2017.8365231.
- [18] D. Tang, B. Qin, X. Feng, and T. Liu, "Effective LSTMs for target-dependent sentiment classification," *arXiv:1512.01100*, 2015, doi: 10.48550/ARXIV.1512.01100.
- [19] C. Wang, S. Huang, and Y. Zhou, "Sentiment analysis of MOOC reviews via ALBERT-BiLSTM model," *MATEC Web Conf.*, vol. 336, p. 05008, 2021, doi: 10.1051/mateconf/202133605008.
- [20] H. Shuqin and R. C. Raga, "A deep learning model for student sentiment analysis on course reviews," *IEEE Access*, vol. 12, pp. 136747–136758, 2024, doi: 10.1109/ACCESS.2024.3463793.
- [21] J. Hwang, "LLMs for sentiment analysis in education: A study in resource-limited settings," *Purdue University Graduate School*, 2025, doi: 10.25394/PGS.28543865.
- [22] C. Sun, L. Huang, and X. Qiu, "Utilizing BERT for aspect-based sentiment analysis via constructing auxiliary sentence," *arXiv:1903.09588*, 2019, doi: 10.48550/ARXIV.1903.09588.
- [23] L. Davoodi, J. Mezei, and M. Heikkilä, "Aspect-based sentiment classification of user reviews to understand customer satisfaction of e-commerce platforms," *Electron. Commer. Res.*, vol. 26, no. 2, pp. 1417–1459, Apr. 2026, doi: 10.1007/s10660-025-09948-4.
- [24] S. Y. Ng, K. M. Lim, C. P. Lee, and J. Y. Lim, "Sentiment analysis using DistilBERT," in *Proc. 2023 IEEE 11th Conf. Syst., Process Control (ICSPC)*, Malacca, Malaysia, 2023, pp. 84–89, doi: 10.1109/ICSPC59664.2023.10420272.
- [25] Z. Fu, Y. C. Hsu, C. S. Chan, C. M. Lau, J. Liu, and P. S. F. Yip, "Efficacy of ChatGPT in Cantonese sentiment analysis: comparative study," *J. Med. Internet Res.*, vol. 26, p. e51069, Jan. 2024, doi: 10.2196/51069.
- [26] N. A. Fauzan, "A zero-shot aspect-based sentiment analysis of public perception toward AI chatbots," *TIERS Inf. Technol. J.*, vol. 6, no. 1, pp. 97–111, Jun. 2025, doi: 10.38043/tiers.v6i1.6488.
- [27] H. Touvron et al., "Llama 2: Open foundation and fine-tuned chat models," *arXiv:2307.09288*, 2023, doi: 10.48550/ARXIV.2307.09288.
- [28] A. Grattafiori, A. Dubey, A. Jauhri, A. Pandey et al., "The Llama 3 herd of models," *arXiv:2407.21783*, 2024, doi: 10.48550/ARXIV.2407.21783.
- [29] K. Schouten and F. Frasinca, "Survey on aspect-level sentiment analysis," *IEEE Trans. Knowl. Data Eng.*, vol. 28, no. 3, pp. 813–830, Mar. 2016, doi: 10.1109/TKDE.2015.2485209.
- [30] G. Zhao, Y. Luo, Q. Chen, and X. Qian, "Aspect-based sentiment analysis via multitask learning for online reviews," *Knowl.-Based Syst.*, vol. 264, p. 110326, Mar. 2023, doi: 10.1016/j.knsys.2023.110326.
- [31] H. Wang et al., "Refining and synthesis: A simple yet effective data augmentation framework for cross-domain aspect-based sentiment analysis," in *Findings Assoc. Computational Linguistics: ACL 2024*, Bangkok, Thailand, 2024, pp. 10318–10329, doi: 10.18653/v1/2024.findings-acl.615.
- [32] B. Warner et al., "Smarter, better, faster, longer: A modern bidirectional encoder for fast, memory efficient, and long context finetuning and inference," *arXiv:2412.13663*, 2024, doi: 10.48550/ARXIV.2412.13663.
- [33] D. Q. Nguyen, T. Vu, and A. T. Nguyen, "BERTweet: A pre-trained language model for English tweets," in *Proc. 2020 Conf. Empirical Methods Natural Language Process.: Syst.*

Demonstrations, 2020, pp. 9–14, doi: 10.18653/v1/2020.emnlp-demos.2.

- [34] A. García-Pablos, M. Cuadros, and G. Rigau, “W2VLDA: Almost unsupervised system for aspect based sentiment analysis,” *Expert Syst. Appl.*, vol. 91, pp. 127–137, Jan. 2018, doi: 10.1016/j.eswa.2017.08.049.
- [35] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, “LoRA: Low-rank adaptation of large language models,” in *Proc. Int. Conf. Learning Representations (ICLR)*, 2022, doi: 10.48550/ARXIV.2106.09685.
- [36] Z. Lan, M. Chen, S. Goodman, K. Gimpel, P. Sharma, and R. Soricut, “ALBERT: A lite BERT for self-supervised learning of language representations,” in *Proc. Int. Conf. Learning Representations (ICLR)*, 2020, doi: 10.48550/ARXIV.1909.11942.